

600-Cell Lattice
(V=120, E=720, F=1200, z=12)



Spectral Traces
 $\text{Tr}(A^2)=2E=1440$, $\text{Tr}(A^3)=6F=7200$



Bare Mode Fraction
 $E/(E+F) = 3/8 = 0.375$ [Theorem 3.1]



Edge Efficiency $\eta = l/R = 1/\phi$
(SSV/PSR metric, Definition 3.2)



$\sin^2\theta_W = \eta \times 3/8 = 3/(8\phi) \approx 0.2318$
(PDG: 0.2312, 0.24%) [Theorem 3.3]



Bond Counting: $\epsilon = 2\sin^2\theta_W/(z+1) = 3/(52\phi)$
Isotropic Shift: $\delta H = \epsilon I_3$ [Steps A-D]



$\cos \theta = -(2/3)(1 + 3/(104\phi))$
 $\theta = 132.731^\circ$ (PDG: 132.732° , 0.003%)
[Theorem 4.3]



Lepton Masses (1 calibration, 0 shape params)
 μ : 105.47 MeV (0.18%) τ : 1774.1 MeV (0.15%)

SM-6: The Complete Derivation Chain